

Name:



Gosford High School

2023 Trial HSC examination

Mathematics Extension 1

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- In Questions 11–14, show relevant mathematical reasoning and/ or calculations

Total Marks

70

Section I – 10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Detach the *Multiple-choice answer sheet* from the last page of this question booklet.

Section II – 60 marks

- Attempt Questions 11–14
- Allow about 1 hours and 45 minutes for this section

Section I

(10 marks)

Attempt Questions 1 – 10

Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for Questions 1 – 10.

1 Which expression is equal to $\int \sin^2(3x) dx$

A. $\frac{1}{2} \left(x - \frac{1}{3} \sin(3x) \right) + C$

B. $\frac{1}{2} \left(x + \frac{1}{3} \sin(3x) \right) + C$

C. $\frac{1}{2} \left(x - \frac{1}{6} \sin(6x) \right) + C$

D. $\frac{1}{2} \left(x + \frac{1}{6} \sin(6x) \right) + C$

2 Which of the following is the solution to $\frac{2}{x-2} < 2$?

A. $x < 2$ or $x > 3$

B. $2 < x < 3$

C. $-2 < x < 3$

D. $-3 < x < 2$

3 If $\sin x = \frac{3}{5}$ and $\frac{\pi}{2} \leq x \leq \pi$, evaluate $\tan 2x$.

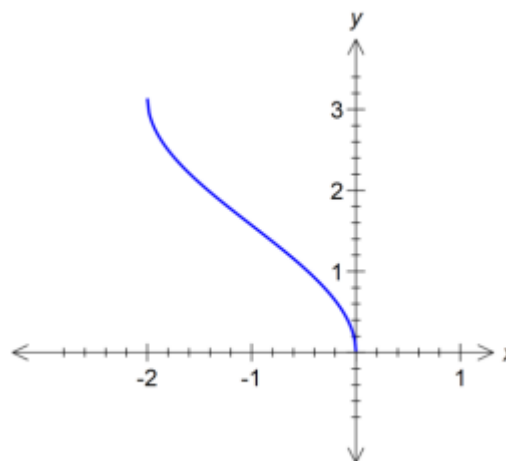
A. $\frac{-7}{24}$

B. $\frac{-24}{7}$

C. $\frac{7}{24}$

D. $\frac{24}{7}$

- 4 Which of the following best describes this function?



- A. $y = \sin^{-1}(x+1)$
- B. $y = \sin^{-1}(x) + 1$
- C. $y = \cos^{-1}(x+1)$
- D. $y = \cos^{-1}(x) + 1$
- 5 In how many ways can 5 people be selected from a group of 6 and then arranged in a line so that the two oldest people in the selected group are at either end of the line? (Note, no two people are the same age)
- A. 720
- B. 144
- C. 72
- D. 36
- 6 The temperature $T^\circ\text{C}$ of water in a jug is given by $T = 20 + 80e^{-0.2t}$. What is the rate at which the water is cooling when its temperature has fallen to half its initial value?
- A. 6°C per minute
- B. 8°C per minute
- C. 12°C per minute
- D. 16°C per minute

7 What is the range of the function $f(x) = \tan^{-1}(\sin x)$

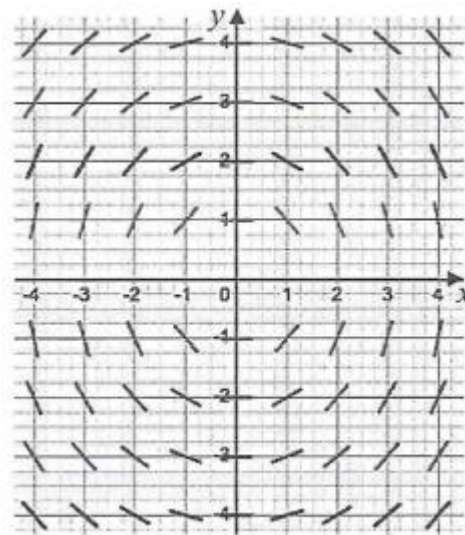
A. $\left(-\frac{\pi}{4}, \frac{\pi}{4}\right)$

B. $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$

C. $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

D. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

8 Which differential equation is represented by the following slope field?



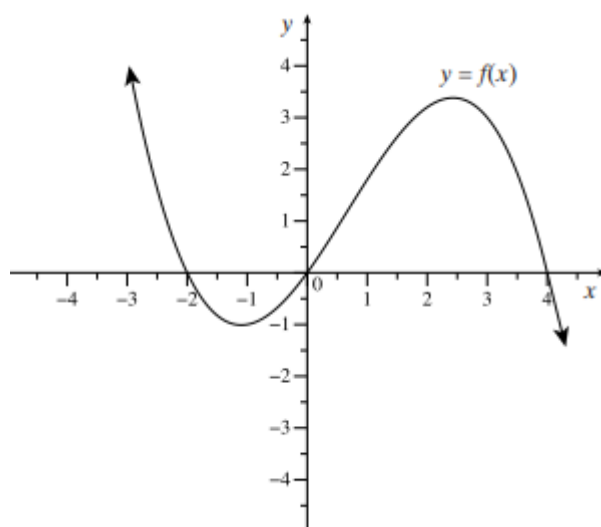
A. $\frac{dy}{dx} = \frac{-x}{y}$

B. $\frac{dy}{dx} = \frac{-y}{x}$

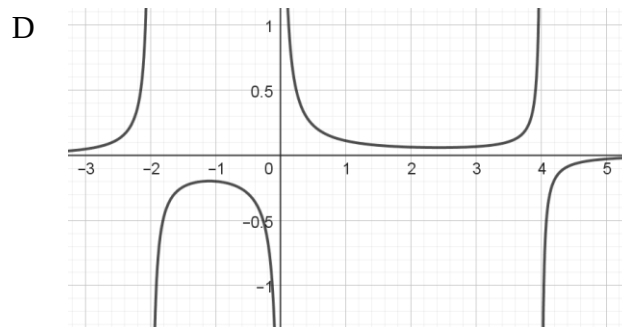
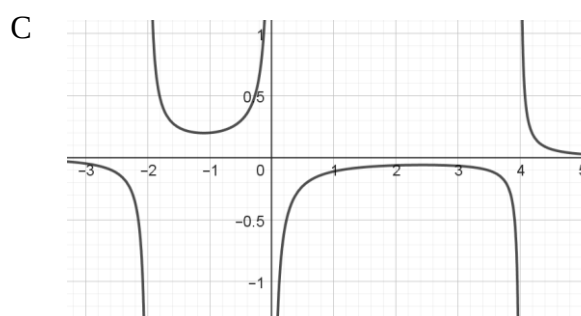
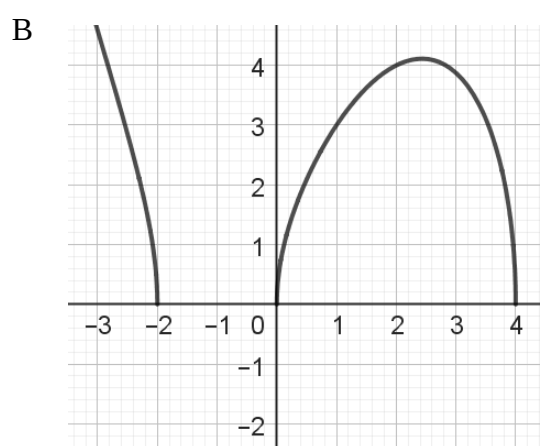
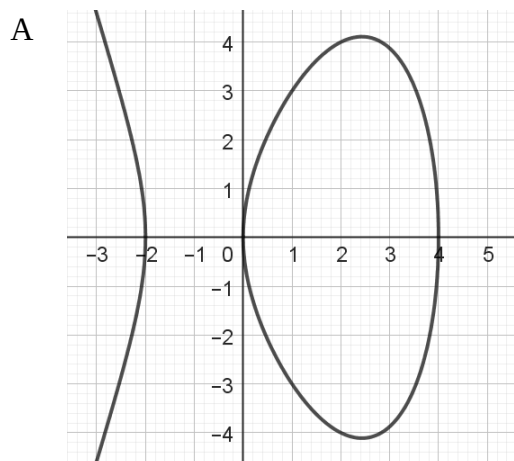
C. $\frac{dy}{dx} = \frac{x}{y}$

D. $\frac{dy}{dx} = \frac{y}{x}$

- 9 The graph of $y = f(x)$ is shown below.



The graph of $y = \frac{1}{f(x)}$ is best represented by:



10 Which of the following is equal to $\cos 3\alpha \sin 2\alpha$?

A. $\frac{1}{2}(\sin 5\alpha + \sin \alpha)$

B. $\frac{1}{2}(\sin 5\alpha - \sin \alpha)$

C. $\frac{1}{2}\left(\sin \frac{5\alpha}{2} + \sin \frac{\alpha}{2}\right)$

D. $\frac{1}{2}\left(\sin \frac{5\alpha}{2} - \sin \frac{\alpha}{2}\right)$

End of Section I

Section II

60 marks

Attempt Questions 11 – 14

Allow about 1 hour and 45 minutes for this section.

Answer each question in a separate booklet.

Question 11 (15 marks) Start a new booklet.

- a) The polynomial $P(x) = 2x^3 - 8x^2 + 7x - 14$ has roots α , $-\alpha$ and β . 1

What is the value of β ?

- b) Use the $t = \tan \frac{x}{2}$ results to show that $\cot x + \tan \frac{x}{2} = \operatorname{cosec} x$ 2

- c) $\int \frac{2}{x^2 + 4} dx$ 2

- d) Find the constant term in the expansion $\left(2x + \frac{3}{x^3}\right)^8$ 3

- e) OAB is a triangle where $\overrightarrow{OA} = 2\hat{i} - k\hat{j}$, $\overrightarrow{OB} = k\hat{i} + 4\hat{j}$ and $AB = 5\sqrt{2}$

- i) Find $\overrightarrow{AB}$ in terms of k . 1

- ii) Find any values of k . 3

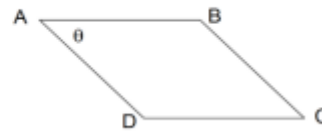
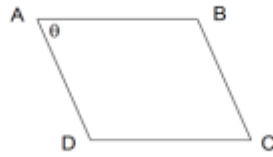
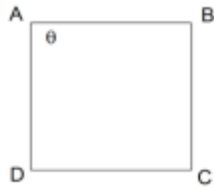
- f) Show that the derivative of $\frac{\cos^{-1}(2x)}{x}$ is equal to $\frac{-\left[2x + \cos^{-1}(2x)\sqrt{1-4x^2}\right]}{x^2\sqrt{1-4x^2}}$ 3

Question 12 (15 marks) Start a new booklet.

- a) Evaluate $\int_0^1 x^2 \sqrt{1+3x^3} dx$ using the substitution $u = 1+3x^3$ 3
- b) Given α, β, γ are the roots of the equation $x^3 - 3x^2 - x + 3 = 0$, evaluate $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$. 3
- c) i) Express $\sqrt{3}\cos\theta - \sin\theta$ in the form $R\cos(\theta + \alpha)$ where $0 \leq \alpha \leq \frac{\pi}{2}$. 2
- ii) Hence, or otherwise, solve $\sqrt{3}\cos\theta - \sin\theta = 1$ for $0 \leq \theta \leq 2\pi$ 2
- d) If $\vec{a} = -6\vec{i} - 2\vec{j}$ and $\vec{b} = \vec{i} + 2\vec{j}$
- i) Show that $\vec{a} \cdot \vec{b} = -10$ 1
- ii) Hence find the vector projection of $\vec{a}$ onto $\vec{b}$ 1
- e) Use mathematical Induction to prove that $11^{2n} + 11^n + 8$ is a multiple of 10 for $n \geq 1$. 3

Question 13 (15 marks) Start a new page.

a)



A square $ABCD$ of side length 1 unit is “gradually” pushed over to become a rhombus. The angle at A θ decreases at a constant rate of 0.1 radian per second?

i) At what rate is the area of the rhombus $ABCD$ decreasing when $\theta = \frac{\pi}{6}$? 3

ii) At what rate is the shorter diagonal of the rhombus $ABCD$ decreasing when $\theta = \frac{\pi}{3}$? 3

b) Show that the solution to the differential equation $\frac{dy}{dx} = 1 + \frac{2y}{3}$, given $y = -1$ when $x = 2$ 4

can be written in the form $y = \frac{e^{\frac{2(x-2)}{3}} - 3}{2}$.

c) A rabbit population grows according to the logistic equation $\frac{dP}{dt} = 0.2P \left(1 - \frac{P}{10000} \right)$, where P is the number of rabbits after t months. The initial population is 1000.

i) Show that $\frac{10000}{P(10000 - P)} = \frac{1}{P} + \frac{1}{10000 - P}$. 1

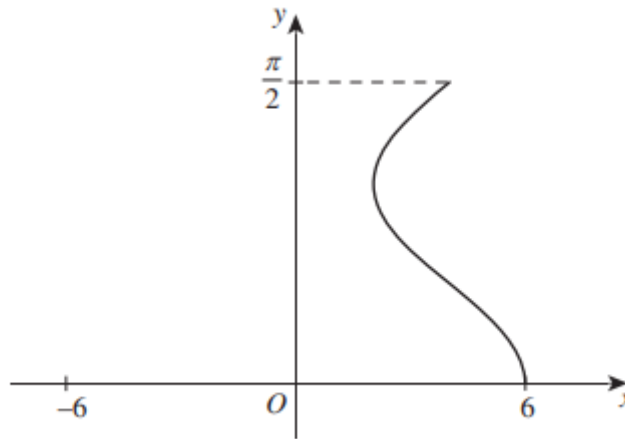
ii) Find the population of rabbits after seven months. 4

Question 14 (15 marks) Start a new booklet

- a) The diagram shows the region bounded by the y-axis and $x = 4\cos(3y) + 2$, where $0 \leq y \leq \frac{\pi}{2}$.

3

The region is rotated about the y-axis to form a solid.



Find the exact volume, V , of the solid formed.

- b) A particle, A, is projected from the origin with an initial velocity of $16\vec{i} + 30\vec{j} \text{ ms}^{-1}$.

At the same time, particle B is projected towards the origin from a point that is 60 m to the right of the origin and 25 m above the origin with an initial velocity of $-8\vec{i} + 20\vec{j} \text{ ms}^{-1}$.

Let the acceleration due to gravity be 9.8ms^{-2} .

The position vector of particle A at time t seconds is given by $A(t) = 16t\vec{i} + (30t - 4.9t^2)\vec{j}$.

(Do NOT prove this.)

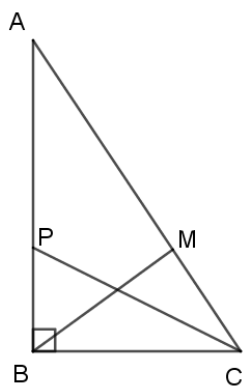
- i) Find the position vector of particle B (i.e. $B(t)$) at time t seconds.

3

- ii) Find the time and point at which the two particles collide.

3

- c) In the figure $\hat{A}BC = 90^\circ$, $PA = 2 \times BP$, $AM = 2 \times MC$.



Let $\overrightarrow{BA} = \underset{\sim}{v}$ and $\overrightarrow{BC} = \underset{\sim}{u}$.

- i) Express $\overrightarrow{BM}$ and $\overrightarrow{CP}$ in terms of $\underset{\sim}{u}$ and $\underset{\sim}{v}$.

3

- ii) If $BM \perp CP$, find $\frac{|BA|}{|BC|}$.

3

End of the Examination. ☺



Gosford High School

2023 Trial HSC examination

Mathematics Extension 1

Solutions



2023 Year 12 HSC Advanced Examination

Section 1: Solutions

- | | | | | |
|-----|-----|-----|-----|-----|
| 1. | A i | B i | C | D i |
| 2. | A | B i | C i | D i |
| 3. | A i | B | C i | D i |
| 4. | A i | B i | C | D i |
| 5. | A i | B i | C | D i |
| 6. | A | B i | C i | D i |
| 7. | A i | B | C i | D i |
| 8. | A | B i | C i | D i |
| 9. | A i | B i | C i | D |
| 10. | A i | B | C i | D i |

Section I

1 Which expression is equal to $\int \sin^2(3x) dx$

C. $\frac{1}{2} \left(x - \frac{1}{6} \sin(6x) \right) + C$

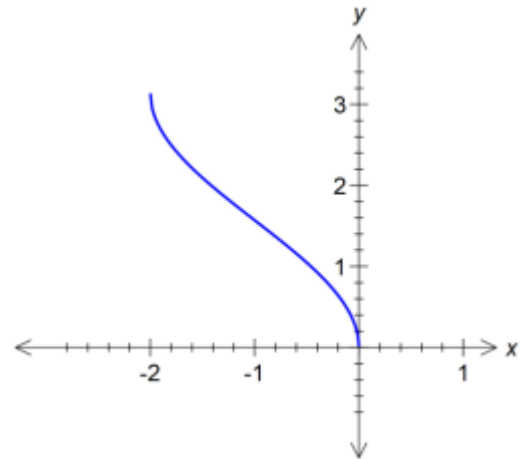
2 Which of the following is the solution to $\frac{2}{x-2} < 2$?

A. $x < 2$ or $x > 3$

3 If $\sin x = \frac{3}{5}$ and $\frac{\pi}{2} \leq x \leq \pi$, evaluate $\tan 2x$.

B. $-\frac{24}{7}$

4 Which of the following best describes this function?



C. $y = \cos^{-1}(x+1)$

5 In how many ways can 5 people be selected from a group of 6 and then arranged in a line so that the two oldest people in the selected group are at either end of the line? (Note, no two people are the same age)

C. 72

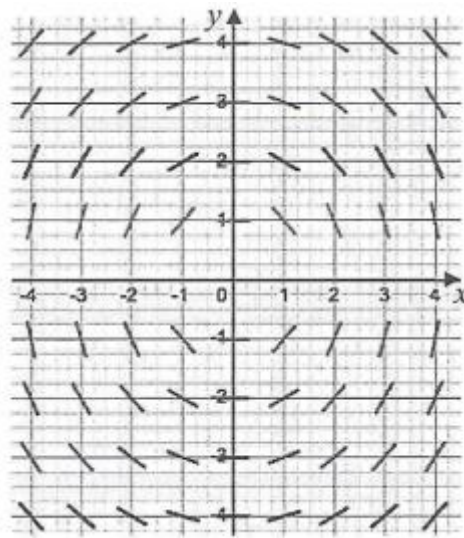
6 The temperature $T^\circ \text{C}$ of water in a jug is given by $T = 20 + 80e^{-0.2t}$. What is the rate at which the water is cooling when its temperature has fallen to half its initial value?

A. 6°C per minute

7 What is the range of the function $f(x) = \tan^{-1}(\sin x)$

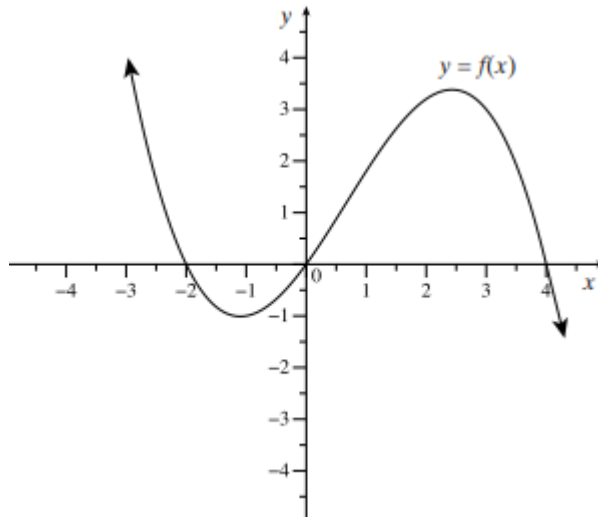
B. $\left[\frac{-\pi}{4}, \frac{\pi}{4} \right]$

8 Which differential equation is represented by the following slope field?

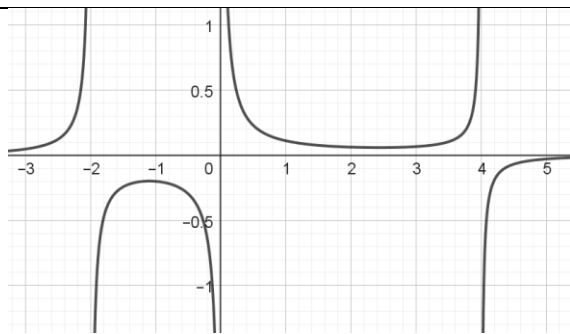


A. $\frac{dy}{dx} = \frac{-x}{y}$

9 The graph of $y = f(x)$ is shown below.



The graph of $y = \frac{1}{f(x)}$ is best represented by:



D.

10 Which of the following is equal to $\cos 3\alpha \sin 2\alpha$?

B. $\frac{1}{2}(\sin 5\alpha - \sin \alpha)$



Mathematics Extension 1

Section II Solutions

Question 11			
a)	The polynomial $P(x) = 2x^3 - 8x^2 + 7x - 14$ has roots α , $-\alpha$ and β . What is the value of β ?		
	Solution: $\alpha - \alpha + \beta = -\frac{(-8)}{2}$ $\therefore \beta = 4$	Marks	Guideline
		1	For correct answer
		Marker's Comment	
			Question done well overall
b)	Use the $t = \tan \frac{x}{2}$ results to show that $\cot x + \tan \frac{x}{2} = \operatorname{cosec} x$		
	Solution: $\text{LHS} = \cot x + \tan \left(\frac{x}{2} \right)$ $= \frac{1-t^2}{2t} + t$ $= \frac{1-t^2+2t^2}{2t}$ $= \frac{1+t^2}{2t}$ $= \operatorname{cosec} x$ $= \text{RHS}$	Marks	Guideline
		2	Correct solution
		1	For correct substitution and making progress towards solution.
		Marker's Comment	Students should take care to only work on one side of the equation at a time. Define the LHS and RHS separately and work on them individually.
c)	$\int \frac{2}{x^2+4} dx$		
	Solution: $\int \frac{2}{x^2+4} dx = 2 \times \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + c$ $= \tan^{-1} \left(\frac{x}{2} \right) + c$	Marks	Guideline
		2	Correct solution
		1	For using inverse tan or getting coefficient of 1.
		Marker's Comment	Question overall done well.

d)	Find the constant term in the expansion $\left(2x + \frac{3}{x^3}\right)^8$											
	Solution: Term in $x^k = \binom{8}{k}(2x)^k(3x^{-3})^{8-k}$ $= \binom{8}{k}2^k \times x^k \times 3^{8-k} \times x^{3k-24}$ $= \binom{8}{k}2^k \times 3^{8-k} \times x^{4k-24}$ For constant term $4k - 24 = 0 \Rightarrow k = 6$ The constant term is $\binom{8}{6} \times 2^6 \times 3^{8-6} = 16128$	<table><tr><th>Marks</th><th>Guideline</th></tr><tr><td>3</td><td>Correct solution</td></tr><tr><td>2</td><td>Correct value for k or correct answer from their k.</td></tr><tr><td>1</td><td>Correct substitution or showing some progress.</td></tr><tr><td>Marker's Comment</td><td>Question overall done well.</td></tr></table>	Marks	Guideline	3	Correct solution	2	Correct value for k or correct answer from their k .	1	Correct substitution or showing some progress.	Marker's Comment	Question overall done well.
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3	Correct solution											
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Marker's Comment	Question overall done well.											
e)	OAB is a triangle where $\vec{OA} = 2\vec{i} - k\vec{j}$, $\vec{OB} = k\vec{i} + 4\vec{j}$ and $AB = 5\sqrt{2}$											
i)	Find $\vec{AB}$ in terms of k .											
	Solution: $\vec{AB} = \vec{OB} - \vec{OA}$ $= k\vec{i} + 4\vec{j} - (2\vec{i} - k\vec{j})$ $= (k-2)\vec{i} + (k+4)\vec{j}$	<table><tr><th>Marks</th><th>Guideline</th></tr><tr><td>1</td><td>For correct answer</td></tr><tr><td>Marker's Comment</td><td>Question done well overall.</td></tr></table>	Marks	Guideline	1	For correct answer	Marker's Comment	Question done well overall.				
Marks	Guideline											
1	For correct answer											
Marker's Comment	Question done well overall.											
ii)	Find any values of k .											
	Solution: $(k-2)^2 + (4+k)^2 = 50$ $\therefore k^2 - 4k + 4 + k^2 + 8k + 16 = 50$ $\therefore k^2 + 2k - 15 = 0$ $\therefore k = 3, -5$	<table><tr><th>Marks</th><th>Guideline</th></tr><tr><td>3</td><td>Correct solution</td></tr><tr><td>2</td><td>Correctly solves their quadratic or gets one of the correct values.</td></tr><tr><td>1</td><td>Showing some progress.</td></tr><tr><td>Marker's Comment</td><td>This question was generally done well if part i was correct.</td></tr></table>	Marks	Guideline	3	Correct solution	2	Correctly solves their quadratic or gets one of the correct values.	1	Showing some progress.	Marker's Comment	This question was generally done well if part i was correct.
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f)	Show that the derivative of $\frac{\cos^{-1}(2x)}{x}$ is equal to $-\frac{[2x + \cos^{-1}(2x)\sqrt{1-4x^2}]}{x^2\sqrt{1-4x^2}}$											
	Solution: $\frac{d}{dx}\left(\frac{\cos^{-1}(2x)}{x}\right) = \frac{x\left(\frac{-2}{\sqrt{1-4x^2}}\right) - \cos^{-1}(2x) \times 1}{x^2}$ $= \frac{\frac{-2x}{\sqrt{1-4x^2}} - \cos^{-1}(2x)}{x^2}$ $= -\frac{[2x + \cos^{-1}(2x)\sqrt{1-4x^2}]}{x^2\sqrt{1-4x^2}}$	<table><tr><th>Marks</th><th>Guideline</th></tr><tr><td>3</td><td>Correct solution</td></tr><tr><td>2</td><td>Correct substitution into Quotient Rule.</td></tr><tr><td>1</td><td>Evidence of Quotient rule or correctly differentiates inverse cos.</td></tr><tr><td>Marker's Comment</td><td>Quotient rule was applied quite well and consistently.</td></tr></table>	Marks	Guideline	3	Correct solution	2	Correct substitution into Quotient Rule.	1	Evidence of Quotient rule or correctly differentiates inverse cos.	Marker's Comment	Quotient rule was applied quite well and consistently.
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3	Correct solution											
2	Correct substitution into Quotient Rule.											
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Marker's Comment	Quotient rule was applied quite well and consistently.											

Question 12

a) Evaluate $\int_0^1 x^2 \sqrt{1+3x^3} dx$ using the substitution $u = 1+3x^3$

Solution:

$$u = 1+3x^3 \Rightarrow \frac{du}{dx} = 9x^2$$

$$\therefore \frac{1}{9} du = x^2 dx$$

At $x = 0, u = 1$

$x = 1, u = 4$

$$\begin{aligned} \int_0^1 x^2 \sqrt{1+3x^3} dx &= \frac{1}{9} \int_1^4 \sqrt{u} du \\ &= \frac{1}{9} \left[\frac{2}{3} u^{3/2} \right] \\ &= \frac{2}{27} [8-1] \\ &= \frac{14}{27} \end{aligned}$$

Marks

Guideline

3

For correct solution

2

Correct integral from substitution or Correct value from their substitution/boundary values. Or equivalent merit.

1

Correct boundary values for u or correct derivative of substitution or equivalent merit.

Marker's Comment

Mostly well done. Main errors occurred when students did not find the bounds for u . Another concerning common mistake was not reading the given u substitution correctly and writing $u = 1 + 3x^2$ making the derivative $u' = 6x$.

b) Given α, β, γ are the roots of the equation $x^3 - 3x^2 - x + 3 = 0$, evaluate $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$

Solution:

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} = \frac{\alpha^2 \beta^2 + \alpha^2 \gamma^2 + \beta^2 \gamma^2}{\alpha^2 \beta^2 \gamma^2}$$

$$\alpha^2 \beta^2 + \alpha^2 \gamma^2 + \beta^2 \gamma^2 = (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2(\alpha^2 \beta\gamma + \alpha\beta^2 \gamma + \alpha\beta\gamma^2)$$

$$= (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2(\alpha\beta\gamma(\alpha + \beta + \gamma))$$

$$= (-1)^2 - 2(-3 \times 3)$$

$$= 19$$

$$\alpha^2 \beta^2 \gamma^2 = (-3)^2$$

$$= 9$$

$$\therefore \frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} = \frac{19}{9}$$

Marks

Guideline

3

Correct solution

2

Correct simplification of sum of the square of the roots, plus 1 correct from 1 Mark or equivalent merit.

1

Correct value for either sum of roots one at a time, two at a time or three at a time.

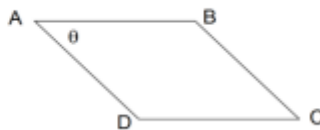
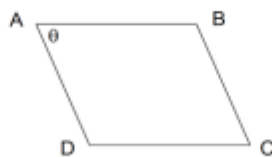
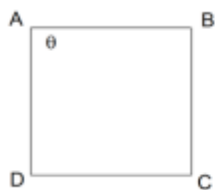
Marker's Comment

Poorly done by most. Better responses were able to successfully simplify

			the fraction and correctly found the values for the sums and products of the roots.	

ci)	Express $\sqrt{3}\cos\theta - \sin\theta$ in the form $R\cos(\theta + \alpha)$ where $0 \leq \alpha \leq \frac{\pi}{2}$												
	Solution: $\sqrt{3}\cos\theta - \sin\theta = R\cos(\theta + \alpha)$ $= R\sin\alpha\cos\theta - R\cos\alpha\sin\theta$ $\therefore R\sin\alpha = \sqrt{3}$ $R\cos\alpha = 1$ $\therefore \tan\alpha = \sqrt{3} \Rightarrow \alpha = \frac{\pi}{6}$ $R = \sqrt{(\sqrt{3})^2 + (-1)^2} = 2$ $\therefore \sqrt{3}\cos\theta - \sin\theta = 2\cos\left(\theta + \frac{\pi}{6}\right)$	<table><tr><th>Marks</th><th>Guideline</th></tr><tr><td>2</td><td>Correct solution</td></tr><tr><td>1</td><td>Correct value for either R or θ</td></tr><tr><td>Marker's Comment</td><td>Well done overall.</td></tr><tr><td></td><td></td></tr></table>	Marks	Guideline	2	Correct solution	1	Correct value for either R or θ	Marker's Comment	Well done overall.			
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Marker's Comment	Well done overall.												
cii)	Hence, or otherwise, solve $\sqrt{3}\cos\theta - \sin\theta = 1$ for $0 \leq \theta \leq 2\pi$												
	Solution: $2\cos\left(\theta + \frac{\pi}{6}\right) = 1 \quad 0 \leq \theta \leq 2\pi$ $\therefore \cos\left(\theta + \frac{\pi}{6}\right) = \frac{1}{2} \quad 0 \leq \theta + \frac{\pi}{6} \leq \frac{13\pi}{6}$ $\therefore \theta + \frac{\pi}{6} = \frac{\pi}{3} \quad \text{or} \quad \frac{5\pi}{3}$ $\therefore \theta = \frac{\pi}{3} - \frac{\pi}{6} \quad \text{or} \quad \frac{5\pi}{3} - \frac{\pi}{6}$ $= \frac{\pi}{6} \quad \text{or} \quad \frac{3\pi}{2}$	<table><tr><th>Marks</th><th>Guideline</th></tr><tr><td>2</td><td>Correct solution</td></tr><tr><td>1</td><td>One correct answer or equivalent merit.</td></tr><tr><td>Marker's Comment</td><td>Mostly well done. Some common errors were only finding one solution.</td></tr><tr><td></td><td></td></tr></table>	Marks	Guideline	2	Correct solution	1	One correct answer or equivalent merit.	Marker's Comment	Mostly well done. Some common errors were only finding one solution.			
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Marker's Comment	Mostly well done. Some common errors were only finding one solution.												
d)	If $\underline{a} = -6\underline{i} - 2\underline{j}$, and $\underline{b} = \underline{i} + 2\underline{j}$												
i)	Show that $\underline{a} \cdot \underline{b} = -10$												
	Solution: $\underline{a} \cdot \underline{b} = -6 \times 1 + -2 \times 2$ $= -10$	<table><tr><th>Marks</th><th>Guideline</th></tr><tr><td>1</td><td>For correct answer</td></tr><tr><td>Marker's Comment</td><td>Very well done.</td></tr><tr><td></td><td></td></tr></table>	Marks	Guideline	1	For correct answer	Marker's Comment	Very well done.					
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Marker's Comment	Very well done.												
ii)	Hence find the vector projection of $\underline{a}$ onto $\underline{b}$												
	Solution: $proj_{\underline{b}} \underline{a} = \frac{\underline{a} \cdot \underline{b}}{\underline{b} \cdot \underline{b}} \underline{b}$ $= \frac{-10}{1+5} (\underline{i} + 2\underline{j})$ $= -2\underline{i} - 4\underline{j}$	<table><tr><th>Marks</th><th>Guideline</th></tr><tr><td>1</td><td>Correct solution</td></tr><tr><td>Marker's Comment</td><td>Mixed results. This is an area the cohort needs to revise as many did not remember the projection vector formula.</td></tr><tr><td></td><td></td></tr></table>	Marks	Guideline	1	Correct solution	Marker's Comment	Mixed results. This is an area the cohort needs to revise as many did not remember the projection vector formula.					
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e)	Use mathematical Induction to prove that $11^{2n} + 11^n + 8$ is a multiple of 10 for $n \geq 1$.													
	Solution: Let $S(n)$ be the statement that $11^{2n} + 11^n + 8$ is divisible by 10 For $n = 1$ $11^2 + 11^1 + 8 = 140$ which is divisible by 10 Hence $S(1)$ is true Suppose there is a k such that $S(k)$ is true. i.e. $11^{2k} + 11^k + 8 = 10M$ for some $M \in \mathbb{Z}^+$ RTP that $11^{2(k+1)} + 11^{k+1} + 8$ is divisible by 10 Now $\begin{aligned} 11^{2(k+1)} + 11^{k+1} + 8 &= 11^{2k} \times 11^2 + 11^k \times 11 + 8 \\ &= 121 \times 11^{2k} + 11 \times 11^k + 8 + 80 - 80 \\ &= 110 \times 11^{2k} - 80 + 11(11^{2k} + 11^k + 8) \\ &= 10(11 \times 11^{2k} - 8) + 11 \times 10M \\ &= 10(11 \times 11^{2k} - 8 + 11M) \\ &= 10N \text{ for some } N \in \mathbb{Z}^+ \end{aligned}$ Since $S(n)$ is true for $n = 1, k$ and $k + 1$ then by the principle of mathematical induction it is true $\forall n \geq 1$.	<table><tr><th>Marks</th><th>Guideline</th></tr><tr><td>3</td><td>Correct solution.</td></tr><tr><td>2</td><td>True for $n=1$ and step 2 or true $n=1$ and some progress towards step 3.</td></tr><tr><td>1</td><td>True for $n=1$ or step 2.</td></tr><tr><td>Marker's Comment</td><td>Mostly well done. Common errors were not defining the pronumeral "M" or incorrectly substituting/simplifying in step 3. Another common error was not fully using $k+1$ in step 3. This highlights that students with incorrect responses need to revise their algebra skills.</td></tr><tr><td></td><td></td></tr></table>	Marks	Guideline	3	Correct solution.	2	True for $n=1$ and step 2 or true $n=1$ and some progress towards step 3.	1	True for $n=1$ or step 2.	Marker's Comment	Mostly well done. Common errors were not defining the pronumeral "M" or incorrectly substituting/simplifying in step 3. Another common error was not fully using $k+1$ in step 3. This highlights that students with incorrect responses need to revise their algebra skills.		
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Question 13
a)


A square $ABCD$ of side length 1 unit is “gradually” pushed over to become a rhombus. The angle at A , θ decreases at a constant rate of 0.1 radian per second?

i)

At what rate is the area of the rhombus $ABCD$ decreasing when $\theta = \frac{\pi}{6}$?

Solution:

$$\begin{aligned} \text{Area} &= \frac{1}{2} AD \times AB \sin \theta + \frac{1}{2} CD \times CB \sin \theta \\ &= \sin \theta \end{aligned}$$

$$\therefore \frac{d\text{Area}}{d\theta} = \cos \theta$$

$$\begin{aligned} \frac{d\text{Area}}{dt} &= \frac{d\text{Area}}{d\theta} \times \frac{d\theta}{dt} \\ &= \cos \theta \times 0.1 \end{aligned}$$

$$\text{When } \theta = \frac{\pi}{6}, \quad \frac{d\text{Area}}{dt} = \frac{\sqrt{3}}{20} \text{ cm}^2 \text{ s}^{-1}$$

Marks
Guideline
3

For correct answer

2

Finds area of rhombus, differentiates correctly and substitutes into function of a function rule. Or equivalent merit.

1

Uses area of a triangle formula or states function of a function rule or equivalent merit.

Marker's Comment
ii)

At what rate is the shorter diagonal of the rhombus $ABCD$ decreasing when $\theta = \frac{\pi}{3}$?

Solution:

From the diagram the shorter diagonal is BD

Let $x = \overline{BD}$

$$\begin{aligned} \therefore x &= \sqrt{1^2 + 1^2 - 2 \times 1 \times 1 \times \cos \theta} \\ &= \sqrt{2 - 2 \cos \theta} \end{aligned}$$

$$= \sqrt{2 - 2 \left(1 - 2 \sin^2 \left(\frac{\theta}{2} \right) \right)}$$

$$= 2 \sin \left(\frac{\theta}{2} \right)$$

$$\frac{dx}{dt} = \frac{dx}{d\theta} \times \frac{d\theta}{dt}$$

$$= 2 \times \frac{1}{2} \cos \left(\frac{\theta}{2} \right) \times 0.1$$

$$\text{When } \theta = \frac{\pi}{3}, \quad \frac{dx}{dt} = \frac{\sqrt{3}}{20} \text{ cms}^{-1}$$

Marks
Guideline
3

Correct solution

2

Correct solution for their shorter diagonal or correct shorter diagonal and its derivative or equivalent merit.

1

Correct expression for shorter diagonal.

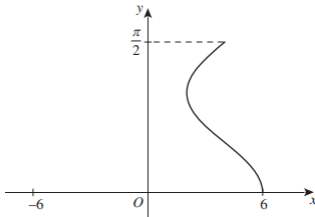
Marker's Comment

b)	Show that the solution to the differential equation $\frac{dy}{dx} = 1 + \frac{2y}{3}$, given $y = -1$ when $x = 2$ can be written in the form $y = \frac{e^{\frac{2(x-2)}{3}} - 3}{2}$.														
Solution: $\frac{dy}{dx} = 1 + \frac{2y}{3}$ $= \frac{3+2y}{3}$ $\frac{dx}{dy} = \frac{3}{3+2y}$ $= \frac{3}{2} \times \frac{2}{3+2y}$ $\therefore x = \frac{3}{2} \ln 3+2y + c$ At $x = 2, y = -1 \Rightarrow 2 = \frac{3}{2} \ln 1 + c \Rightarrow c = 2$ $\therefore x = \frac{3}{2} \ln 3+2y + 2$ $\therefore \ln 3+2y = \frac{2(x-2)}{3}$ $\therefore 3+2y = e^{\frac{2(x-2)}{3}}$ $\therefore y = \frac{e^{\frac{2(x-2)}{3}} - 3}{2}$	<table border="1"> <thead> <tr> <th>Marks</th><th>Guideline</th></tr> </thead> <tbody> <tr> <td>4</td><td>Correct solution</td></tr> <tr> <td>3</td><td>Correct indefinite integral and value of c or equivalent merit.</td></tr> <tr> <td>2</td><td>Correct indefinite integral or equivalent merit.</td></tr> <tr> <td>1</td><td>Re-arranges equation into a usable form or calculates their value of c or equivalent merit.</td></tr> <tr> <td>Marker's Comment</td><td></td></tr> <tr> <td></td><td></td></tr> </tbody> </table>	Marks	Guideline	4	Correct solution	3	Correct indefinite integral and value of c or equivalent merit.	2	Correct indefinite integral or equivalent merit.	1	Re-arranges equation into a usable form or calculates their value of c or equivalent merit.	Marker's Comment			
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Marker's Comment															

c)	A rabbit population grows according to the logistic equation $\frac{dP}{dt} = 0.2P\left(1 - \frac{P}{10000}\right)$, where P is the number of rabbits after t months. The initial population is 1000.																
i)	Show that $\frac{10000}{P(10000 - P)} = \frac{1}{P} + \frac{1}{10000 - P}$																
	Solution: $\begin{aligned} \text{RHS} &= \frac{1}{P} + \frac{1}{10000 - P} \\ &= \frac{10000 - P + P}{P(10000 - P)} \\ &= \frac{10000}{P(10000 - P)} \\ &= \text{LHS} \end{aligned}$	<table><tr><th>Marks</th><th>Guideline</th></tr><tr><td>1</td><td>Correct solution</td></tr><tr><td>Marker's Comment</td><td></td></tr><tr><td></td><td></td></tr></table>	Marks	Guideline	1	Correct solution	Marker's Comment										
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1	Correct solution																
Marker's Comment																	
ii)	Find the population of rabbits after seven months.																
	Solution: $\begin{aligned} \frac{dP}{dt} &= 0.2P\left(1 - \frac{P}{10000}\right) \\ 5 \frac{dP}{dt} &= P\left(\frac{10000 - P}{10000}\right) \\ \frac{1}{5} \frac{dt}{dP} &= \frac{10000}{P(10000 - P)} \\ &= \frac{1}{P} + \frac{1}{10000 - P} \\ \therefore \frac{1}{5} t &= \ln P - \ln 10000 - P + c \\ \text{At } t = 0, P = 1000 &\Rightarrow 0 = \ln 1000 - \ln 9000 + c \Rightarrow c = \ln 9 \\ \therefore \frac{t}{5} &= \ln P - \ln(10000 - P) + \ln 9 \\ &= \ln\left(\frac{9P}{10000 - P}\right) \\ \text{At } t = 7, \frac{7}{5} &= \ln\left(\frac{9P}{10000 - P}\right) \\ \therefore \frac{9P}{10000 - P} &= e^{7/5} \\ \therefore 9P + Pe^{7/5} &= 10000e^{7/5} \\ \therefore P &= \frac{10000e^{7/5}}{9 + e^{7/5}} \\ &\approx 3106.19 \end{aligned}$ The population after seven months is 3106.	<table><tr><th>Marks</th><th>Guideline</th></tr><tr><td>4</td><td>For correct answer</td></tr><tr><td>3</td><td>Uses value of t to find correct exponential equation or equivalent merit.</td></tr><tr><td>2</td><td>Correct indefinite integral and value of c or equivalent merit.</td></tr><tr><td>1</td><td>Re-arranges equation into a usable form or calculates their value of c or equivalent merit.</td></tr><tr><td>Marker's Comment</td><td></td></tr><tr><td></td><td></td></tr></table>	Marks	Guideline	4	For correct answer	3	Uses value of t to find correct exponential equation or equivalent merit.	2	Correct indefinite integral and value of c or equivalent merit.	1	Re-arranges equation into a usable form or calculates their value of c or equivalent merit.	Marker's Comment				
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Marker's Comment																	

Question 14

- a) The diagram shows the region bounded by the y-axis and $x = 4\cos(3y) + 2$, where $0 \leq y \leq \frac{\pi}{2}$.
The region is rotated about the y-axis to form a solid.



Find the exact volume, V , of the solid formed.

Solution:

$$\begin{aligned}
 V &= \pi \int_0^{\pi/2} x^2 dy \\
 &= \pi \int_0^{\pi/2} (16\cos^2(3y) + 16\cos(3y) + 4) dy \\
 &= \pi \int_0^{\pi/2} (8 + 8\cos(6y) + 16\cos(3y) + 4) dy \\
 &= \pi \left[12y + \frac{8}{6}\sin(6y) + \frac{16}{3}\sin(3y) \right]_0^{\pi/2} \\
 &= \pi \left[6\pi + \frac{4}{3}\sin 3\pi + \frac{16}{3}\sin \frac{3\pi}{2} - 0 \right] \\
 &= \pi \left(\frac{18\pi - 16}{3} \right) \text{ units}^3
 \end{aligned}$$

Marks	Guideline
3	For correct answer
2	Correct integral or correct solution from their integral involving a $\cos^2$ term. Or equivalent merit.
1	Correct expression for x^2 or correct substitution for their $\cos^2$ expression or correct solution for their integral or equivalent merit.
Marker's Comment	Generally well done. Some students missed the $16\cos(3y)$ or changed the $16\cos^2(3y)$ incorrectly.

- b) A particle, A , is projected from the origin with an initial velocity of $16\mathbf{i} + 30\mathbf{j} \text{ ms}^{-1}$.
At the same time, particle B is projected towards the origin from a point that is 60 m to the right of the origin and 25 m above the origin with an initial velocity of $-8\mathbf{i} + 20\mathbf{j} \text{ ms}^{-1}$.
Let the acceleration due to gravity be 9.8 ms^{-2} .
The position vector of particle A at time t seconds is given by $A(t) = 16t\mathbf{i} + (30t - 4.9t^2)\mathbf{j}$.

- i) Find the position vector of particle B (i.e. $B(t)$) at time t seconds.

Solution:

For particle B

$$\begin{aligned}
 \ddot{y} &= -g & \ddot{x} &= 0 \\
 \dot{y} &= -gt + 20 & \dot{x} &= -8 \\
 y &= -\frac{g}{2}t^2 + 20t + 25 & x &= -8t + 60 \\
 \therefore B(t) &= (-8t + 60)\mathbf{i} + (25 + 20t - 4.9t^2)\mathbf{j}
 \end{aligned}$$

Marks	Guideline
3	Correct solution
2	One correct component and progress towards the other component.
1	One correct component.
Marker's Comment	Well done, although some students had $x = -8t$ and $y = -\frac{g}{2}t^2 + 20t$

ii)	Find the time and point at which the two particles collide.													
	Solution: The particles collide when $A(t) = B(t)$ $\therefore 16t\hat{i} + (30t - 4.9t^2)\hat{j} = (-8t + 60)\hat{i} + (25 + 20t - 4.9t^2)\hat{j}$ Comparing $\hat{j}$ components: $-8t + 60 = 16t \Rightarrow t = 2.5$ $A(2.5) = 16 \times 2.5\hat{i} + (30 \times 2.5 - 4.9(2.5)^2)\hat{j}$ $= 40\hat{i} + \frac{355}{8}\hat{j}$ The particles collide after 2.5 s at a point 40 m to the right and 44.375 m above the origin	<table><tr><th>Marks</th><th>Guideline</th></tr><tr><td>3</td><td>Correct solution</td></tr><tr><td>2</td><td>Finds the time at which the two particles collide or equivalent merit.</td></tr><tr><td>1</td><td>Recognises we need to make $A(t) = B(t)$.</td></tr><tr><td>Marker's Comment</td><td>Well done</td></tr><tr><td></td><td></td></tr></table>	Marks	Guideline	3	Correct solution	2	Finds the time at which the two particles collide or equivalent merit.	1	Recognises we need to make $A(t) = B(t)$.	Marker's Comment	Well done		
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Marker's Comment	Well done													
c)	In the figure $\angle ABC = 90^\circ$, $PA = 2 \times BP$, $AM = 2 \times MC$.  Let $\overrightarrow{BA} = \underline{v}$ and $\overrightarrow{BC} = \underline{u}$.													
i)	$\overrightarrow{BM}$ and $\overrightarrow{CP}$ in terms of $\underline{u}$ and $\underline{v}$.													
	Solution: Now $\overrightarrow{CA} = \overrightarrow{BA} - \overrightarrow{BC} = \underline{v} - \underline{u}$ $\therefore \overrightarrow{CM} = \frac{1}{3}(\underline{v} - \underline{u}) \text{ and } \therefore \overrightarrow{BP} = \frac{1}{3}\underline{v}$ $\overrightarrow{BM} = \overrightarrow{BC} + \overrightarrow{CM} \qquad \overrightarrow{CP} = \overrightarrow{CB} + \overrightarrow{BP}$ $= \underline{u} + \frac{1}{3}(\underline{v} - \underline{u}) \qquad = -\underline{u} + \frac{1}{3}\underline{v}$ $= \frac{2}{3}\underline{u} + \frac{1}{3}\underline{v}$	<table><tr><th>Marks</th><th>Guideline</th></tr><tr><td>3</td><td>Correct solution</td></tr><tr><td>2</td><td>Finds either BM or CP.</td></tr><tr><td>1</td><td>Finds CM or BP.</td></tr><tr><td>Marker's Comment</td><td>Many, many students had the vectors in the wrong direction e.g., $\overrightarrow{CP} = \overrightarrow{BC} + \overrightarrow{BP} = \underline{u} + \frac{\underline{v}}{3}$ Some had $\frac{1}{2}$ instead of $\frac{1}{3}$</td></tr></table>	Marks	Guideline	3	Correct solution	2	Finds either BM or CP .	1	Finds CM or BP .	Marker's Comment	Many, many students had the vectors in the wrong direction e.g., $\overrightarrow{CP} = \overrightarrow{BC} + \overrightarrow{BP} = \underline{u} + \frac{\underline{v}}{3}$ Some had $\frac{1}{2}$ instead of $\frac{1}{3}$		
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ii)	If $BM \perp CP$, find $\frac{ BA }{ BC }$													
	Solution: Now $\overrightarrow{BM} \cdot \overrightarrow{CP} = 0 \Rightarrow \left(\frac{2}{3}\underline{u} + \frac{1}{3}\underline{v}\right) \cdot \left(-\underline{u} + \frac{1}{3}\underline{v}\right) = 0$ $\therefore -\frac{2}{3}(\underline{u} \cdot \underline{u}) + \frac{2}{9}(\underline{u} \cdot \underline{v}) - \frac{1}{3}(\underline{v} \cdot \underline{u}) + \frac{1}{9}(\underline{v} \cdot \underline{v}) = 0$ $\therefore \frac{1}{9}(\underline{v} \cdot \underline{v}) - \frac{2}{3}(\underline{u} \cdot \underline{u}) = 0 \text{ as } (\underline{u} \cdot \underline{v} = 0)$ $\therefore \frac{\underline{v} \cdot \underline{v}}{\underline{u} \cdot \underline{u}} = 6$ $\therefore \frac{ \underline{v} ^2}{ \underline{u} ^2} = 6 \Rightarrow \frac{ BA }{ BC } = \frac{ \underline{v} }{ \underline{u} } = \sqrt{6}$	<table><tr><th>Marks</th><th>Guideline</th></tr><tr><td>3</td><td>For correct answer</td></tr><tr><td>2</td><td>Uses $\underline{u} \cdot \underline{v} = 0$ to simplify equation or equivalent merit.</td></tr><tr><td>1</td><td>Uses $BM \cdot CP = 0$.</td></tr><tr><td>Marker's Comment</td><td>Generally well done, although there were some issues with expanding the dot product.</td></tr></table>	Marks	Guideline	3	For correct answer	2	Uses $\underline{u} \cdot \underline{v} = 0$ to simplify equation or equivalent merit.	1	Uses $BM \cdot CP = 0$.	Marker's Comment	Generally well done, although there were some issues with expanding the dot product.		
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